BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 182 · Recursion: call stack and tracing

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. For Factorial(3), list the successive input values before any returns occur.

Q2. What is the difference between a function calling another function and returning a value to its caller?

Q3. A stack contains bottom to top P, Q, R. After one pop, what is at the top?

Q4. A function Square(x) returns x*x. Find Square(2)+Square(3), showing both calls.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

stack frame — An entry containing the state for one active function call.

unwind — Return from calls in reverse order after a base case is reached.

local variable — A variable owned by a particular invocation.

call stack — Last-in, first-out storage for active calls.

base case — A condition that ends recursion.

New learning

Build the stack — SumTo(4) waits for SumTo(3), which waits for SumTo(2), which waits for SumTo(1).

Separate the local values — Each stack frame holds its own n: 4, 3, 2 or 1. The values do not overwrite one another.

Unwind from the top — SumTo(1) returns 1; SumTo(2) returns 3; SumTo(3) returns 6; SumTo(4) returns 10.

Worked example

Draw the call-stack frames for SumTo(3). Which call returns first?

1. Push — The frames are SumTo(3), SumTo(2), then SumTo(1), with SumTo(1) at the top.

2. Stop — SumTo(1) is the base case; it returns 1 first.

3. Unwind — SumTo(2)=2+1=3, followed by SumTo(3)=3+3=6.

We Do · Practise with guidance

Trace SumTo(5). Label the top of the stack immediately before the first return.

5 → 4 → 3 → 2 → 1

Predict the next step before revealing the explanation.

Call sequence

SumTo(5) → SumTo(4) → SumTo(3) → SumTo(2) → SumTo(1).

Top frame

SumTo(1) is on top immediately before the first return.

Returns

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1, 3, 6, 10, 15. SumTo(5) returns 15.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

List the call-stack frames from bottom to top for SumTo(4).

Question 2

Trace the returned values for SumTo(6).

Question 3

A pupil writes that SumTo(4) returns 4+3+2+1 before SumTo(1) has executed. Explain the order error.

Exit Ticket · Answer before checking

Every recursive call uses the same local-variable storage.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 3, 2, 1, 0. Each call reduces n by one until the base case.

Q2. A call transfers control into a function; return finishes that invocation and gives its result to the caller.

Q3. Q. The topmost R is removed first.

Q4. Square(2)=4 and Square(3)=9; the total is 13.

You Do pages · Independent questions

Q1. SumTo(4), SumTo(3), SumTo(2), SumTo(1). SumTo(1) is the top frame.

Q2. 1, 3, 6, 10, 15, 21 in the order calls return; the outer function returns 21.

Q3. The expression describes the final arithmetic but not execution order. The deepest call SumTo(1) must return first; only then can waiting calls compute their sums.

Exit Ticket · Model answer

False. Each active call has its own stack frame and local values.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: